THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

DODOMA REGION
REGIONAL FORM IV MOCK EXAMINATION – MAY 2019

033

BIOLOGY - 1

TIME: 2.30 Hours **Tuesday, 7th May 2019, a.m.**

INSTRUCTIONS

1. This paper consists of section A, B and C with a total of thirteen (13) questions
2. Answer all questions in section A and B and one question from section C.
3. Except for diagrams that must be drawn in pencil, all writing should be in blue or black ink.
4. Calculators, cellular phones and any unauthorized materials are not allowed in the examination room.
5. Write your examination number on every page of your answer booklet(s)

This paper consists of **6** printed pages

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SECTION A: (20 MARKS)

Answer all questions in this section

1. For each of the items (i)-(x), choose the correct answer from among the given alternatives and write its letter beside the item number in the answer booklet.

(i) Which of the following is a non-living part of a cell?

A. Cell wall and nucleus
B. Cell wall
C. Cytoplasm and endoplasmic reticulum
D. Plant cell

( )

(ii) The following is a list of diseases caused by bacteria;

A. Pneumonia, elephantiasis, cholera
B. Pneumonia, malaria, cholera
C. Bilharzia, cholera, pneumonia
D. Tuberculosis, pneumonia, small pox
E. Tuberculosis, pneumonia, cholera

( )

(iii) The phylum chordata includes all animals that have

A. Mammary gland
B. Post anal tail
C. A backbone
D. Scales
E. Warm dry skin

( )

(iv) By which process does carbon dioxide pass from the blood into the alveoli of the lungs?

A. Diffusion
B. Osmosis
C. Respiration
D. Transpiration
E. Oxidation

( )

(v) If a parent cell has 12 chromosomes, how many chromosomes will be in the daughter cells formed after meiosis?

A. Six (6)
B. Twelve (12)
C. Eighteen (18)
D. Twenty (20)
E. Thirty six (36)

( )

(vi) Which of the following tissues is meristematic?

A. Cornified layer of the skin
B. Collenchyma
C. Cambium
D. Sclerenchyma
E. Xylem

( )

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(vii) Which substance in tobacco combines with haemoglobin to prevent the haemoglobin from carrying oxygen?

A. Carbon dioxide
B. Carbon monoxide
C. Nicotine
D. Tar
E. Cellulose fibre

( )

(viii) A bean plant can bear either terminal or axial flowers. When a terminal flowered plant (T) is pollinated with an axial flowered plant (t), the offspring produced were 210 terminal flowered and 200 axial flowered. Which of the following represents the genotype of the parents?

A. TT x Tt
B. Tt x Tt
C. TT x tt
D. Tt x tt
E. T x t

( )

(ix) Which of the following is a metabolic waste product?

A. Tears
B. Saliva
C. Mucus
D. Faeces
E. Urine

( )

(x) The theory of special creation postulates that:

A. Organisms emerged from non living matter
B. Organisms arose from pre-existing forms
C. A supernatural being created the earth and all the living things
D. Life started after elements joined to form organic compounds
E. Mutation change the cause of evolution

( )

2. Match the responses in list B with the word or phrases in list A by writing the letter of the correct response beside the item number

List A	List B
(i) A state of physical and mental well being	A. Beri beri
(ii) Attack and engulf foreign organisms	B. Vector
(iii) Cholera and meningitis	C. Good manner
(iv) Vitamin B deficiency	D. Pandemic
(v) Spreads diseases without being affected	E. Natural immunity
(vi) Chemicals on the surface of pathogens	F. Antibodies
(vii) Ready-made antibodies	G. Bacterial
(viii) Removes pathogens	H. Lymphocytes

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(ix) Behaving in an acceptable way (x) The outbreak of disease across whole continent	I. Personal hygiene J. Immunity K. Phagocytes L. Protozoans M. Scurvy N. Parasites O. Antigen P. Sneezing Q. Healthy R. Epidemic S. Yawning T. Passive immunity

SECTION B (60 MARKS)

Answer all questions in this section

3. (a) Outline four (4) features, which distinguish a biology laboratory from other school facilities.
   (b) Outline four steps involved in using a microscope.

4. (a) Give two reasons why biologist classify organisms?
   (b) The following organisms belong to the kingdom Fungi: Mushroom, mucor, yeast and bread mould
   (i) Group the organisms according to their respective phyla
   (ii) Write down two (2) characteristics that all fungi have in common.

5. (a) define the following terms
   (i) Ecosystem
   (ii) Producers
   (iii) Decomposers
   (b) (i) Differentiate the term "food chain" from a "trophic level"
   (ii) Construct a food chain by using organisms named in the following list: Grass, lions, Zebra and Bacteria

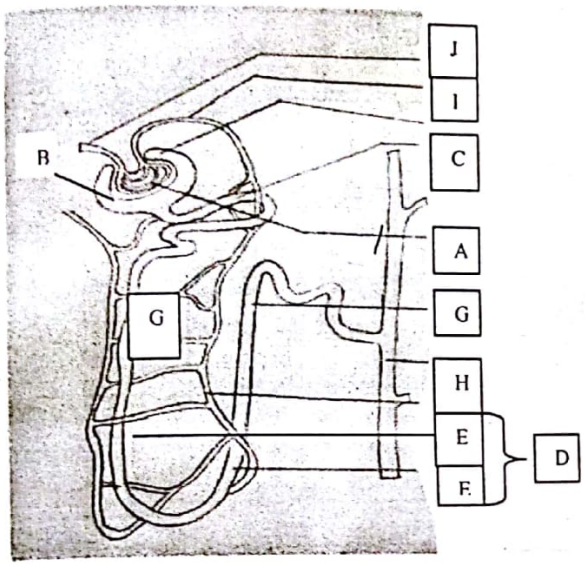
6. (a) (i) What is Mitosis?
   (ii) State four significance of mitosis
   (b) Explain briefly the types of seed germination with one example in each.

7. (a) Give the meaning of the following terminologies as used in genetics.
   (i) Sex linked genes
   (ii) Sex determination
   (iii) Phenotype

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(b) Juma and Rosa are newly married couple expecting a baby. Juma is carrier for albinism while Rosa is normal. Workout the genotype of their probable first born by using genetic cross diagram

8. Figure 1 below shows different parts of a nephron



(a) Name the structures labeled A to J
(b) Where does filtration occur in this structure?
(c) How does fluid in part C differ from that in H?
(d) Where are Na and Cl ions mainly reabsorbed?
(e) Where glucose is mainly reabsorbed?
(f) What would happen to an organism if structure D is relatively longer or short?

9. (a) Name the Hormone, which control the following processes in Human body
(i) Development of female secondary sexual characteristics.
(ii) Preparation of the uterus for pregnancy
(iii) Production and secretion of milk by mammary gland.
(iv) Contraction of uterus wall during birth.

(b) State the functions of the following hormone
(i) Follicle stimulating hormone
(ii) Lutenizing hormone
(iii) Androgens

10. (a) (i) Differentiate sexual reproduction from asexual reproduction with one example in each
(ii) Give four advantages of sexual reproduction

(b) Briefly explain the mechanism of double fertilization in flowering plants .

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SECTION C (20 MARKS)

Answer one (1) question from this section

11. Explain five factors which contribute to irresponsible sexual behavior among young people and their effect in the community.

12. With the aid of a well labeled diagram discuss the mechanism of hearing in human being.

13. Describe four evidences of organic evolution.

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REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

DODOMA REGION
REGIONAL FORM IV MOCK EXAMINATION – MAY 2019

013/1 **GEOGRAPHY - 1**

TIME: 3 Hours **Monday, 6th May 2019 p.m.**

INSTRUCTIONS

1. This paper consist of sections **A, B, C** and **D** with a total of twelve (12) questions.
2. Answer all questions in sections **A, B** and **C**, and one (1) question from each part of section **D**.
3. Map extract of **NANSIO (Sheet 22/1)** is provided.
4. Credit will be given for the use of relevant diagrams.
5. Calculators, Cellular Phones and any other unauthorized materials are not allowed in the examination room.
6. Write your **examination number** on every page of your answer booklet(s).

This paper consists **5** printed pages

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SECTION A: (25 MARKS)

Answer all questions in this section

PHYSICAL AND MATHEMATICAL GEOGRAPHY

1. For each of the items (i) – (x) choose the correct answer from among the given alternatives and write its letter against the item number in the answer booklet provided.

(i) Which of the following activities does not involve geographical knowledge?
(A) Weather forecasting
(B) Map making and interpretation ( )
(C) Planning for the location of agricultural activities
(D) Quarrying
(E) Religious matters

(ii) Spinning of a body on its axis.
(A) Solar eclipse
(B) Lunar eclipse
(C) Revolution ( )
(D) Rotation
(E) Axis

(iii) Physical geography is the branch of geography which is concerned with _____
(A) The study of land formation processes, weather and climate
(B) The study of human activities on the earth's surface
(C) The study of field work, photograph work and statistics ( )
(D) Environment that include settlements and economic activities
(E) The study of Solar system

(iv) What is the approximate difference in meridian time between Geita (33° E) and St. Carol (80° E0?
(A) 4 hours
(B) 3 hours
(C) 2 hours
(D) 6 hours ( )
(E) 5 hours

(v) In which of the following is a Savanna type of climate?
(A) East Africa
(B) Central Asia
(C) West Africa
(D) South Central Africa ( )
(E) Far East

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(vi) One of the following is not a type of rock.
(A) Sedimentary rock
(B) Igneous rock
(C) Limestone ( )
(D) Metamorphic rock
(E) Sedimentary, Igneous and Metamorphic rocks

(vii) The following are processes of chemical weathering.
(A) Solution, carbonation, hydrolysis, hydration and oxidation
(B) Hydrolysis, siltation, solutions and mass wasting
(C) Exfoliation, solution, chemical form and oxidation ( )
(D) Transportation, hydrolysis, hydration and solution
(E) Oxidation, weathering and solution

(viii) In which stage of the river deposition is dominant?
(A) The young stage
(B) The middle stage
(C) Long profile stage ( )
(D) The old stage.
(E) None of above

(ix) This is among the important conditions for a beach to form.
(A) Strong swash and weak back swash
(B) Steep sloping shore
(C) A coastline lying parallel to the prevailing winds ( )
(D) Weak swash.
(E) Strong swash

(x) Earthquakes may not occur due to:
(A) Collision of plate tectonics
(B) Large scale farming activities ( )
(C) Lateral and radial forces
(D) Shaking of earth's surface
(E) Situations of faults and volcanic activities

2. Match the words / phrases from list A with their corresponding statements in list B

LIST A	LIST B
(i) Solar system	A. Is a stream or a river that flow into large stream or main stem river or a lake
(ii) Tributaries	B. The sun together with all planets and other bodies that revolve around it
(iii) Distributaries	C. All planets and other bodies revolving around the sun
(iv) Cirque	D. Long, narrow, winding ridges of sand and gravels
(v) Eskers	

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	E. River profile that has been renewed F. Is a stream that flow away from a main stream channel G. Is the steep sided, semicircular basin curved on the sides of a mountain by glacier

3. (a) Define the term Soil erosion.
   (c) State major ways of controlling soil erosion to your environment.

SECTION B: (27 MARKS)

Answer all questions in this section

APPLICATION OF STATISTICS, INTRODUCTION TO RESEARCH AND ELEMENTARY SURVEYING

4. Carefully study the following information in the table below that shows the number of students at KWANEEMA SECONDARY SCHOOL and then answer the questions that follows:

FORM I	FORM II	FORM III	FORM IV
60	125	90	35

(a) Present the number of students in each class in percentage
(b) Draw a pie graph and represent the data provided
(c) State two merits and two demerits of the method above.

5. (a) What is research?
   (b) What are the qualities of a good research?

6. (a) Define the term leveling.
   (b) Briefly explain the usefulness of levelling techniques
   (c) Explain three categories of errors that are likely to occur in surveying.

SECTION C: (27 MARKS)

MAP READING AND PHOTOGRAPHY INTERPRETATIONS.

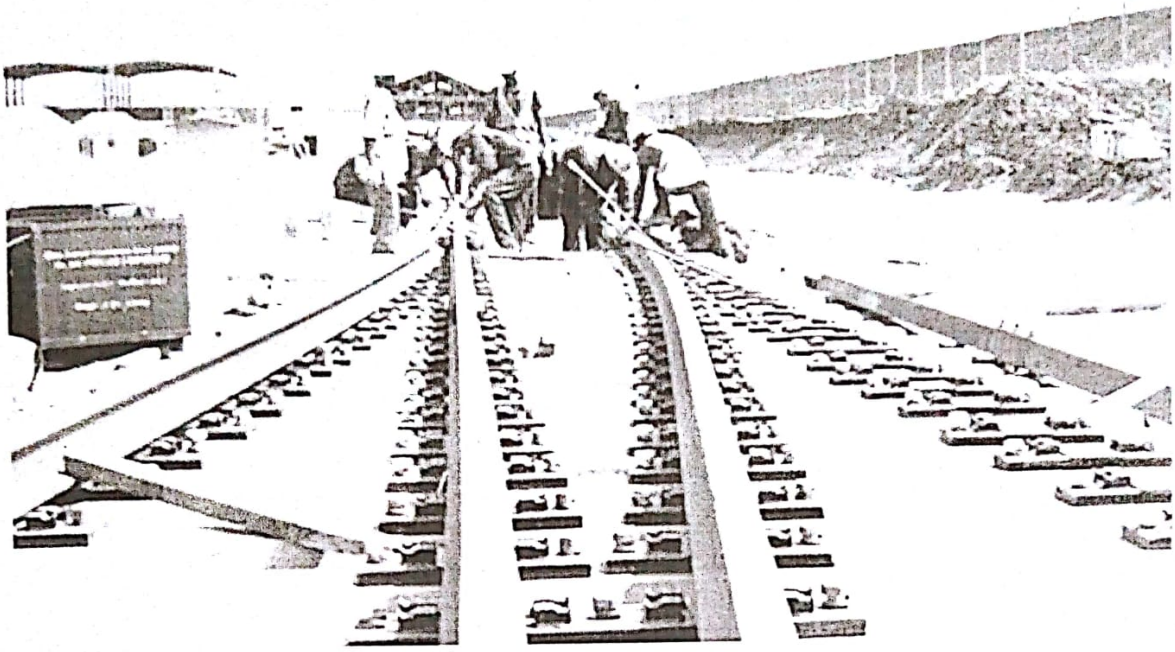
7. Carefully study the Map Extract of **NANSIO** (Sheet 22/1) provided then answer the questions which follow.
   (a) Calculate the distance of a line connecting grid reference 180646 and 216670
   (b) Determine the area covered by Lake Victoria from South to Western part of the map in kilometer square.
   (c) State any four possible economic activities on the area.
   (d) Describe the relief of the mapped area.
   (e) State the contour interval of the area.
   (f) Point out ways used to express scale on the map.

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8. Study the following photography and answer the questions that follows



(i) What type of photograph is shown?

(ii) Give two reasons to support your answer in (i) above

(iii) What type of activity is taking place in the photograph provided?

(iv) Give three advantages of type of transport shown in the photograph.

(v) Mention at least two regions in Tanzania which depends on railway to transport goods and passengers.

SECTION D (20 MARKS)

Answer two questions from this section, one question from each part.

PART I: REGIONAL FOCAL STUDIES.

9. Identify and explain the problems facing fishing industry in East Africa.

10. Discuss on the advantages of developing manufacturing sector in Tanzania.

PART II: ENVIRONMENTAL ISSUES, POPULATION AND SETTLEMENT.

11. Discuss three major population problems in developing countries like Tanzania.

12. Analyze six measures toward environmental conservations.

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DODOMA REGION
REGIONAL FORM IV MOCK EXAMINATION – MAY 2019

032/2A **CHEMISTRY – 2A**
(ACTUAL PRACTICAL)

TIME: 3 Hours **Thursday, 25th April 2019 a.m.**

INSTRUCTIONS

1. This paper consists of three questions. Answer ALL Questions
2. Question 1 carries twenty (20) marks the rest carry fifteen (15) marks each
3. Qualitative analysis guidance pamphlets may be used after a thorough check by the supervisor
4. Cellular phones and calculators are not allowed in the examination room.
5. You may use the following constants

Atomic masses:

H=1, S=32, O=16, Na=23, Cl=35.5 C=12

1 litre=1dm^3=1000cm^3

This paper consists of 4 printed pages

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1. You are provided with the following solutions;

E: Solution containing 3.65g of Hydrochloric acid per dm^3 of solution.

K: Solution containing 7.15g of pure hydrated sodium carbonate, $Na_2CO_3.XH_2O$ per $0.25dm^3$ of solution.

L: Methyl orange indicator solution

Procedure

(a) Fill the burette with solution **E**.

(b) Pipette 20 cm^3 or 25 cm^3 of solution **K** into a titration flask. Add few drops of **L**. Titrate this base solution against the acid solution until an end point is reached.

(c) Record your results in a table form

Summary

(a) The volume of pipette used was_____cm^3

(b) The colour change at the end point was from _____ to _____The volume of acid solution E needed for complete neutralization was___cm^3

Questions

(a) Write down the balanced chemical equation for the reaction.

(b) Calculate the

(i) Molarity of acid solution **E**

(ii) Molarity of base solution **K**

(iii) Find the value of water of crystallization in $Na_2CO_3.XH_2O$.

2. You are provided with the following:

Z: A Solution containing $32.24g/dm^3$ of $Na_2S_2O_3.5H_2O$

U: A solution containing 2M HCl

T: Distilled water

V: Stop watch/clock

Procedure

(a) Write letter ***X*** on a piece of paper with black ink

(b) Measure 50 cm^3 of **Z** and pour it into a conical flask

(c) Measure $10cm^3$ of **U** by using different measuring cylinder and pour it into the same conical flask immediately start the stop watch and observes the changes from above.

(d) Stand the conical flask on the paper above the cross. Look down vertically through the mouth of the conical flask and make sure you can see the cross at the cross at the bottom of the flask.

(e) Swirl the flask while keep on observing until the letter **X** disappears.

(f) Record the time taken for disappearance of letter **X**

(g) Repeat the whole procedure using $40cm^3$, $30cm^3$,$20cm^3$ and finally $10cm^3$

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Table of results

Experiment	Volume of solutions used , cm^3			Time (t)	Rate of eaction, $\frac{1}{t}$ in s^{-1}
	Z	U	T		
1	50	10	0		
2	40	10	10		
3	30	10	20		
4	20	10	30		
5	10	10	40		

Questions

a) Complete filling the table.
b) Write the balance chemical equation between Z and U
c) Plot the graph of concentration of sodium thiosulphate against Rate 1/t.
d) What substance was produced during the reaction which obscured the mark X
e) What conclusion can you draw from the result of this experiments
f) Briefly explain the smell and colour observed as the reaction take place

Sample **Q** is a simple salt containing one action and one anion .Carefully carry out all he experiments described in table 2. Record all your observation and appropriate nferences to identify the ions present in Q

S/N	Experiments	Observation	Inference
1	Observe sample **Q**		
2	Put a spatulafull of sample **Q** in a test tube and add distilled water.		
3	Transfer 0.5g of sample **Q** in a test tube,add dilute HCl		
4	Transfer 0.5g of sample **Q** in a test tube,add concentrated sulphuric acid		
5	Dissolve sample **Q** then divide the resulting solution into three portions (i) To the first portion add sodium hydroxide solution		

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	(ii) To the second portion add ammonia solution (iii) To the third portion add potassium ferricyanide solution{$K_3Fe(CN)_6$} (iv) To the fourth portion add lead acetate solution followed by acetic acid solution.		

Conclusion

(a) The cation in sample Q is----------

(b) The anion in sample Q is----------

(c) The compound Q is----------

(d) State two chemical properties of a metal in Q

(e) State two uses of Q

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DODOMA REGION
REGIONAL FORM IV MOCK EXAMINATION – MAY 2019

041/1 **BASIC MATHEMATICS**

TIME: 3 Hours **Monday, 6th May 2019 a.m.**

INSTRUCTIONS

1. This paper consists of section A and B with a total of sixteen questions.
2. Answer all questions in section A and any four (4) questions from section B. each question in section B carries ten (marks).
3. All necessary working and answers for each question done must be shown clearly
4. Mathematical table and graph papers may be used
5. Calculators, cellular phones and any unauthorized material are not allowed in the examination.
6. Write your examination number on every page of your answer booklet(s)

This paper consists of 4 printed pages

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SECTION A (60 MARKS)

Answer ALL questions in this section.

1. (a) Express 0.05473
   (i) Correct to (3) significant figure
   (ii) Correct to (3) decimal places
   (iii) In standard form

   (b) Change 0.835835 ..... into simple fraction.

2. (a) (i) Evaluate the following without using tables: ½ log64 + log50 – log 4

   (ii) Solve for x given that $2^{2x+3} + 1 = 9(2^x)$, if $y = 2^x$

   (b) If $x = 2+\sqrt{3}$ find the value of $x^2 + \frac{1}{x^2}$

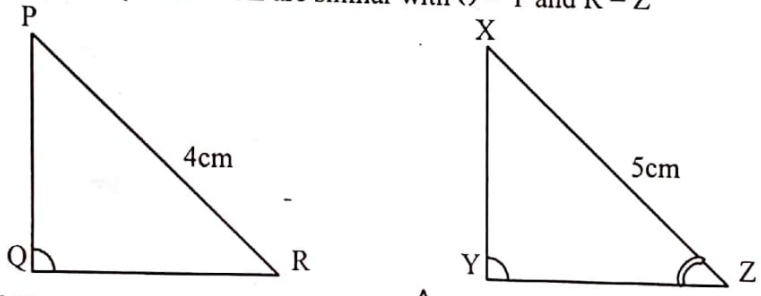
3. (a) If a*b – ab + b, find x given that $(x*2)*5 = 60$

   (b) In an interview it was found that out of 30 households, 10 have a car but not television, 13 have television but not a car and 4 have neither car nor television. Using Venn diagram illustrate that how many household have both car and television.

4 (a) Given: $\underline{a} = 3i + 2j$, $\underline{b} = 4i + 2j$ and $\underline{c} = 6i - 12j$. Find the scalar m and n such that $m\underline{a} + n\underline{b} = \underline{c}$

   (b) If the line from (4, y) to (x_2, -3) bisect at (1, -1) find the value of y and x_2.

5. (a) Triangle PQR and XYZ are similar with $\hat{O} = \acute{Y}$ and $R = \check{Z}$



   If PR = 4cm, XZ = 5cm and area of $\triangle$XYZ = 1.25cm2. Find the area of $\triangle$PQR

   (b) A regular octagon having an area of 240cm^2 is inscribed in a circle. Find the radius of a circle.

6. (a) 9 workers working 8 hours a day complete a piece of work in 52 days. How will it take for 13 workers to complete the same job by working 6 hours a day?

   (b) A variable Z varies directly as x and inversely as y^2. If Z = 4.5 when x = 6 and y=2, find Z when x = 3.4 and y = ½

7. (a) A mathematics text book costs Tsh. 5,500/-. The writer of the book gets 10% of the price of each book sold. How much will the writer get if 16,500 copies are solid?

   (b) Find the time which Tsh. 30,000/- will earn Tsh. 6,000/- at the rate of 5% interest per annum.

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8. (a) The 10th term of AP is 42 more than the 7th term of the same sequence. If the first term is -14, find the 2nd term of the sequence.

(b) Mr. Lewis deposited Tsh. 400,000/- in a bank at a compound interest 5% semiannually for 4 years. Calculate the amount he received at the end of the period.

9. (a) Given that in triangle ABC, AB = 8cm, AC =6cm and the degree measured of the angle BAC is 60°, Calculate
(i) The length of BC using cosine rule
(ii) The degree measure of angle ABC by using cosine rule.

(b) The distance from the top of the cliff to the boat is 200km and the distance from the boat to the foot of the cliff is 100 km. Calculate the angle of elevation from the boat to the bank.

10. (a) Solve the equation $x^2 + x - 2 = 0$ by completing the square method.

(b) Find the two consecutive odd numbers whose product is 195.

SECTION B (40 MARKS)

11. Joseph wishes to buy up to 40 notebooks for his stationary. He can buy either type A for 3000/- each or type B for 6,000/- each. He has a total of 150,000/- to spend and he must have at least 10 notebooks of type A and at least 5 notebooks of type B in his stock.

(a) Write down all inequalities which represent the given information.

(b) If he makes a profit of Tsh. 400/- on each notebook of type A and a profit of 1,000/- on each notebook of type B. How many notebooks of each type he should buy to maximize profit?

12. The frequency distribution table below shows the scores of 100 students in a Mathematics test.

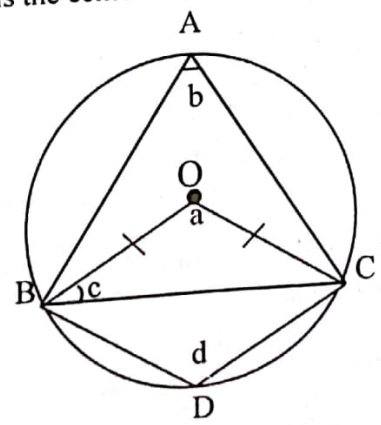
Scores	10-19	20-29	30-39	40-49	50-59	60-69	70-79	80-89	90-99
No of Students	2	6	14	18	20	16	12	8	4

(a) Use assumed mean A = 54.5 to calculate the mean score

(b) Draw cumulative frequency curve, use the graph to estimate median.

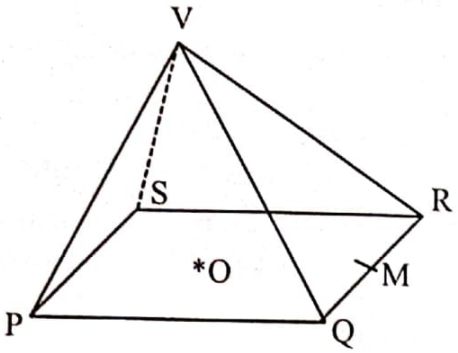
(c) Write down the modal class.

13. (a) In the figure below, O is the center of a circle and angle $a = 140^{o}$, find angle b, c and d.



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(b) The figure below represents a right pyramid with vertex V and a rectangular base PQRS. VP = VQ = VR = VS = 18cm. PQ = 16 cm and QR = 12cm. M and Q are the mid points of QR and PR respectively.



(i) Find the size of angle between VP and the plane PQRS
(ii) Calculate the volume of the pyramid

14. (a) Prepare the balance sheet for the balances given in the table below

Capital	45,000
Drawings	28,000
Creditors	4,300
Stock	5,000
Debtors	8,000
Premises	16,000
Motor Vehicle	8,000
Bank	4,000
Cash	9,000
Net profit	2,700
Bank overdraft	6,000
Loan from bank	20,000

(b) From the balance sheet constructed in (a) above, determine the following
(i) Working capital (ii) Current ratio

15. (a) Solve by using matrix method

$$\begin{Bmatrix} 4x + 2y = 40 \\ x + 3y = 35 \end{Bmatrix}$$

(b) (i) Find t given that $A = \begin{pmatrix} 6 & 2t \\ t & 0 \end{pmatrix}$ has no inverse

(ii) A point (x, y) is rotated through 90° and then reflected about the line y = -x, find
(a) A single matrix for this double transaction
(b) The image of a point (6, 3) under this double transaction.

16. (a) Given the function $f(x) = x^2 + 6x - 7$. Find
(i) The minimum point of $f(x)$
(ii) x and y intercept of $f(x)$
(iv) The axis of symmetry of $f(x)$
(v) The range of $f(x)$

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DODOMA REGION
REGIONAL FORM IV MOCK EXAMINATION – MAY 2019

031/2A **PHYSICS 2A**
(ACTUAL PRACTICAL)

TIME: 2.30 Hours **Monday, 29th April 2019 a.m.**

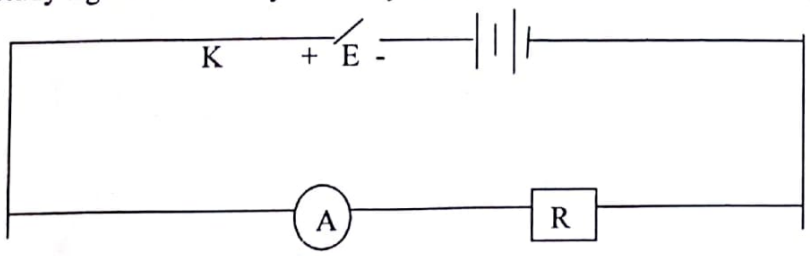
INSTRUCTIONS

1. This paper consists of two (2) questions
2. Answer all the questions
3. Where calculations are involved show your work clearly
4. Each questions carries 25 marks
5. Calculators cellular phones and any unauthorized materials are not allowed in the examination room
6. Use $\pi = 3.14$

This paper consist of 2 printed pages

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1. Study figure below very carefully and set the circuit based on tools provided



Procedure

(a) With R = 2Ω close the switch k and record the reading I from the ammeter.

Repeat the procedure above from value R = 3Ω, 4Ω, 5Ω, 6Ω, each time record the ammeter readings

(b) Record the results in a tabular from including a columns of 1/I

(c) Plot a graph of R against 1/I

(d) Find the slope s of your graph

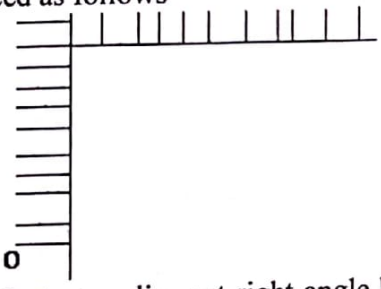
(e) Use equation E = I(R + r) determine the value of E of the cell

(f) What was the aim of the experiment?

2. In this experiment you are provided with two plane mirrors one optical pin, drawing paper soft board and protractor and thumb pins.

Proceed as follows

θ



(a) Draw two liner at right angle by using your rule and protractor on the plane sheet of paper

(b) Place the sheet of paper securely fixed on the drawing board using the thumb pin

(c) Place one mirror on each line on the sheet of paper

(d) Fix an optical pin 0 and look into both mirrors and count the number of image when θ = 90°

(e) Repeat the procedure above for θ = 60°, 45°, 36° and 30 such that you obtain four more reading

(f) Tabulate your result for the value of θ, n and $\frac{360°}{\theta}$

(g) (i) Plot a graph of number of image (n) of against $\frac{360°}{\theta}$

(ii) Determine the slope of the graph

(iii) Find the value of n- intercept from the graph

(iv) Using the graph value of slope, n-intercept and $\frac{360°}{\theta}$ derive the equation relating the number of images and $\frac{360°}{\theta}$

(v) (i) What is the aim of experiment

(ii) What happen to the number of image as the value of angle θ is reduced

(iii) What happen to the number of image when θ = 0°?

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DODOMA REGION
REGIONAL FORM IV MOCK EXAMINATION – MAY 2019

031/2A **PHYSICS 2A**

24 HOURS ADVANCED INSTRUCTION

TIME: 3 Hours

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1. **QUESTION NUMBER ONE**

   i. Two dry cells size D 1.5V each National Panasonic
   ii. One ammeter (0-3A)
   iii. One resistor box capable of supplying resistance of 1ohms up to 10 ohms
   iv. 8 pieces of connecting wires
   v. One key

2. **QUESTION NUMBER TWO**

   i. Two plane mirror with holders
   ii. One optical pin
   iii. One soft drawing board
   iv. One protector
   v. Four thumb pins
   vi. A 30cm ruler

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THE UNITED REPUBLIC OF TANZANIA

PRESIDENTS OFFICE REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

DODOMA REGION FORM IV MOCK EXAMINATIONS

MAY 2019

032/2A CHEMISTRY 2A

24HRS ADVANCE INSTRUCTIONS

1.0 IMPORTANT

GREAT CARE MUST BE TAKEN NOT TO DIVULGE THESE INSTRUCTIONS TO BOTH CANDIDATES AND UNAUTHORIZED PERSONS EITHER DIRECTLY OR INDIRECTLY.

1.2:.NOTE THAT YOU WILL NOT BE ALLOWED TO OPEN THE ENVELOPE CONTAINING THE QUESTION PAPERS BEFORE THE TIME OF COMMENCEMENT OF THE EXAMINATION

2.0 PREPARATION AND LABELLING OF CHEMICALS

2.1 Question 1

Prepare the following

- 0.1M Hydrochloric acid solution, label it **E** provide 100cm3 per candidate
- 0.05M Hydrated Sodium carbonate label it **K** Provide 100cm3 per candidate
- Provide methyl orange indicator for sharing in the ratio of one to two candidates Label it **L**

2.2 Question 2

Prepare the following

- 0.2M Sodium thiosulphate($Na_2S_2O_3$)solution label it **Z**
- 2M Dilute hydrochloric acid solution (HCl) label it **U**
- Distilled water and allow 100cm^3 per candidate label it **T**
- Provide one stopwatch, and a white plain paper per candidate.

2.3 Question 3.

- Provide 3g of Iron(II) Sulphate label it **Q**
- Provide enough distilled water 300cm^3 per candidate.
- Heat source or burner(candidates must not exceed four for sharing)

NOTE:**TO EXAMINATION SUPERVISOR AND LABORATORY TECHNICIAN/HEAD OF CHEMISTRY DEPARTMENT**

The Laboratory technician or head of chemistry department should perform an experiment of question 1 and 2 during the last 30 minutes of the examination time and submit to REO the experimental results together with the candidates' data

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(70)

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DODOMA REGION
REGIONAL FORM IV MOCK EXAMINATION – MAY 2019

011 **CIVICS**

TIME: 3 Hours **Friday, 3rd May 2019, p.m.**

INSTRUCTIONS

1. This paper consists of sections A, B and C.
2. Answer ALL questions in section A and B and three (3) questions from section C
3. Section A and B carry 20 marks each and section C carries 60 marks
4. Cellular phones are not allowed in the examination room.
5. Write your Examination number on every page of your answer booklet (s)

This paper consists of 4 printed pages

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SECTION A (20 MARKS)

1. For each of the following item (i) -(x) choose the most correct answers among the given alternative and write its letter besides the item number.

(i) Tanzania constitution has the following function of local government except.
- (a) To perform the function of local government within its area.
- (b) Paying required taxes, licenses, and other duties
- (c) To ensure enforcement of law and public safety.
- (d) To consolidate democracy within its area.

(ii) The president of Tanzania has the following powers as stated by constitution.
- (a) The president as a prerogative mercy
- (b) Overseeing discipline of the parliament
- (c) The head of government Business in parliament
- (d) To make the decision on the annual budget.

(iii) The direct democracy was possible in ancient Greece because of the following reason except?
- (a) They were respecting human rights
- (b) The size of the population was small
- (c) The geographical areas were also small
- (d) There was discrimination against some of citizen.

(iv) The judiciary organ of the state plays one of the following function
- (a) Policy formulation
- (b) Budget formulation through the executive
- (c) The process of interpreting laws to the people.
- (d) It enforces policies formulated.

(v) A person with good leadership skills will enable the society to get the:
- (a) Limitation of resources utilization available in a particular area
- (b) Minority decision dominates all the decision of the majority.
- (c) People will be well informed on changes that occurring day to day.
- (d) It promotes single headed job than group work.

(vi) Per capita income is obtained by dividing the total annual income of the country by
- (a) Labour force employed in Agriculture
- (b) Export factor
- (c) Energy consumption
- (d) Total population of the country.

(vii) Which of the following factors is an indicator of social development?
- (a) Life expectancy of the people (b) Labour
- (c) Capital (d) Good government

(viii) Which of the following Institution is not dealing with loan provision?
- (a) Bureau de change
- (b) The National Social Security Fund (NSSF)
- (c) The banks
- (d) The National Insurance Corporation (NIC)

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ix) One of the following statements is unclear about Hygienic environment.
(a) keep water bodies and source clean
(b) preserving purity of an air
(c) factory should not be far from residential area to secure the market
(d) Sewage systems to remain perfect.

x) Poverty caused mainly by lack of resources and capabilities to acquire basic needs. Which one is an indicator of poverty?
(a) No external political influence
(b) High life expectancy
(c) Low per capital income
(d) They politically stable.

Match the items in list A with the correct response in list B by writing the letter of the corresponding response beside the item number in the answer booklet(s) provided:-

LIST A	LIST B
(i) Psychological treatment to people with emotional or mental suffering	A. Empathy
(ii) The ability to make the best choice out of many available options	B. Customs
(iii) Ability to think and come up with new idea and ways of doing things	C. Values
(iv) Discussing and looking for solution	D. Arts
(v) Feeling compassion to other peoples problem	E. Archives
(vi) Pattern of conduct shared by the people in a given community but always change with time	F. Taboos
(vii) Physical artifacts such as furniture made by human beings	G. Myths
(viii) Practices that persisted in a society for a long time and do not change	H. Recreation
(ix) Relate to activities that people do after working hours for relaxation and enjoyment	I. Decision making skills
(x) A system of communication in speech or writing used by a particular community	J. Interview
	K. Norms
	L. Critical thinking
	M. Mediation
	N. Craft
	O. Problem solving skills
	P. Guidance and counseling
	Q. Language

SECTION B: (20 MARKS)

Read the following passage carefully and answer the questions which follow

Trees play an important role in preserving and protecting our environment of Earth. They absorb carbon dioxide and release oxygen into the air, in this way they help to reduce the amount of pollution in the Earth's atmosphere. They also play an important part in the water cycle as they release moisture into the atmosphere through their leaves. This helps the formation of clouds which bring rain. Loss of trees can therefore be a cause of drought.

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In addition, at times when there is a lot of rain, trees can soak up much of the extra water and so they help to prevent soil erosion as they work like nets trapping soil and stopping it from being washed away. In addition fallen threes, branches and leaves decay on the ground, making the soil more fertile.

An individual tree is home to all kinds of animals; insects, birds, mammals, reptiles as well as other plants. In a forest, millions of types of trees can be found. For example tropical rain forest, which occupy less than 8% of the Earth's land area, are home to well over 50% of all Land plants and animals.

In addition to these benefits, products from trees provide us with thousands of everyday items. Many fruits and seeds are eaten by humans and animals. Then there is wood, people do thousands of things with wood. It is used in the construction of both traditional and modern buildings. Inside most buildings there are furniture like wardrobes, tables, chairs, beds and so on, all of which are made- up of wood. In the streets there, may be all kinds of structures are made-up of wood. Signboards and telegraph poles are just too. Wood is also an important source of fuel.

Questions:

(a) Provide a suitable title for the passage
(b) Mention four (4) importance of trees to human beings and animals
(c) What do you think are the products from trees (two products)
(d) Write down the role played by trees on environment (2 roles)
(e) In your own words as form four student what are your opinions to the people who destruct trees in their environment (2 points)

4. (a) Briefly explain five elements of culture
   (b) Outline five merits of making responsible decision

SECTION C: (60 MARKS)

Answer three (3) questions from this section. Each question carries 20 marks

5. Accidents caused by drivers have now reached the point of intolerable proportions. As a Tanzanian, suggests six measures to alleviate road accidents.
6. You cannot separate life and work. Examine the importance of work to human development by giving six points.
7. Examine six root causes of poverty in Tanzania.
8. "Culture is the back born of any society". Discuss six importance of culture in your community.
9. Examine six basic principles of democracy to a country like Tanzania.
10. Show six possible solutions to the challenges of globalization to developing country like Tanzania.

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DODOMA REGION
REGIONAL FORM IV MOCK EXAMINATION – MAY 2019

033/2A

BIOLOGY 2A
(ACTUAL PRACTICAL)

TIME: 2.30 Hours **Wednesday, 24th April 2019, a.m.**

INSTRUCTIONS

1. This paper consists of two (2) questions. Answer all questions.
2. Each question carries 25 marks
3. Except for diagrams that must be drawn in pencil, all writings should be in blue or black ink ball pen.
4. Calculators, cellular phones and any unauthorized materials are not allowed in the examination room.
5. Write your examination number on every page of your answer booklet(s)

This paper consists of **2** printed pages

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1. You have been provided with solution R.

(a) Perform experiment using the reagents provided to identify the type of food substance(s) present in the solution R. Tabulate your result as shown in Table 1.

Table 1.

Food tested	Procedures	Observation	Inference

(i) Explain the role(s) of each food substance identified in 1(a)

(ii) Give the name of the end product after digestion for each food substance identified in 1(a)

(iii) One of the identified food substances above is very important to children for growth. Give its name.

(iv) Name the malnutritional disorder associated with lack of food substance named in 1(d) above

2.Study specimen O, P, Q, X, Y and Z.

(i) Identify specimen O, P, Q, X, Y and Z by using their common names

(ii) (a) State the phyla and class for each specimens O, P and Q.

(b) Mention four observable characteristics of specimen P.

(iii) State the divisions for each specimen X, Y and Z.

(iv) Use a knife or Razor blade and cut a specimen Z longitudinally, then draw and label the longitudinal view of the specimen Z.

(v) Give four differences between specimen X and specimen Y.

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OFISI YA RAIS
TAWALA ZA MIKOA NA SERIKALI ZA MITAA

MKOA WA DODOMA
MTIHANI WA UTAMILIFU, KIDATO CHA NNE 2019

021 **KISWAHILI**

Muda: Saa 3 **Ijumaa, 3 Mei 2019, asubuhi.**

MAELEKEZO

1. Karatasi hii ina sehemu **A, B, C, D** na **E**.
2. Jibu maswali yote katika sehemu A, B na D Swali moja (1) kutoka sehemu **C** na maswali matatu (3) kutoka sehemu **E**. Swali la (15) ni la lazima.
3. Zingatia maagizo ya kila sehemu ya kila swali.
4. **Simu za mikononi** haziruhusiwi ndani ya chumba cha mtihani.
5. **Vikokotozi** haviruhusiwi ndani ya chumba cha mtihani.
6. Andika namba yako ya mtihani katika kila ukurasa wa kijitabu chako cha kujibia.

Mtihani huu una kurasa 4 zilizochapwa

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SEHEMU A: (ALAMA 10)
UFAHAMU
Jibu maswali YOTE katika sehemu hii

1. Soma kifungu cha habari kifuatacho kisha jibu maswali yanayofuata.

Hapo zamani za kale kulikuwa na mvuvi ambaye aliishi na mkewe katika kibanda kibovu na ufukwe wa bahari. Siku moja wakati akivua samaki kwa ndoano, alimwopoa samaki mmoja mkubwa sana wa ajabu!

Samaki Yule alimrairai mvuvi huku akilia na kusema, "Tafadhali, tafadhali, naomba uniache niishi. Mimi si samaki bali ni mwana wa mfalme, nirudishe baharini na uniruhusu niende"

Kusikia vile, yule mvuvi akajibu, "Ohoo, usiseme maneno mengi, sina haja na samaki anayezungumza, kwa hiyo, nenda zako mapema iwezekanavyo." Akamnasua na kumrudisha baharini. Aliporudi nyumbani na kumweleza mkewe kuhusu mkasa ule, mkewe alikuja juu na kumuuliza mumewe, kwa nini ulimwachia bila kumwomba lolote mwana yule wa mfalme?

"Hapana" alijibu yule mvuvi, "kwani ningeomba nini? Akauliza. Kusikia hivyo, mkewe alijibu kwa ghadhabu "Kweli wewe hawayani, sisi tunaishi katika kijumba hiki kibovu, kwa nini hukumwomba atusaidie nyumba nzuri?"

Yule mvuvi alirudi mbio ufukweni na kuanza kuita kwa sauti! "Ewe Bwana wa bahari, njoo unisikilize, mke wangu amenituma nije nikuombe msaada wa nyumba"

Mara yule samaki akajitokeza na kumwamuru arejee mara moja nyumbani. Mvuvi alipofika nyumbani alimkuta mkewe katika nyumba nzuri ndogo, iliyokuwa na bustani nzuri ya maua yenye rihi ya kupendeza na matunda matamu, huku kuku na bata wakirandaranda huku na kule. Furaha ikapasua anga na nyuso zikajaa bashasha.

Lakini baada ya majuma mawili mke wa mvuvi alianza kunyong'onyea na siku moja akapasua jipu. "Kajumba haka hakatoshi na hata kabustani tuliko nako ni kadogo." Upumbavu gani huu? Nenda basi kamwambie yule samaki atupatie Kasri.

Baada ya kusikia hivyo, mvuvi alimsihi sana mkewe ili waridhike na walichokuwa nacho wasiende tena kuomba kwa samaki kwani lisingekuwa jambo la busara. Lakini mkewe alimshinikiza aende, hivyo mvuvi akaenda shingo upande. Alishangaa kuona kuwa samaki aliliridhia ombi la mkewe wakapata kasri kubwa lililokuwa na watumishi wa kila fani na bustani kubwa iliyojaa kila aina ya wanyama na ndege. Mvuvi akaamini kwamba yeye na mkewe wataishi kwa furaha kwa siku zote za maisha yao.

Walipoamka, mkewe akaja na ombi jipya, alimtaka mumewe akamwombe samaki ili wao ndio wawe mfalme na malkia wa nchi ile. Baada ya kumwona mumewe anasita, akamwambia "Kama wewe hutaki kuwa mfalme, mimi nitakuwa mfalme nenda kamwambie huyo samaki wako." Samaki alipopewa ombi hilo alijibu kwa maneno machache "Mtaka yote kwa pupa hukosa yote."

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Maswali

(i) Ni jipu gani mke wa mvuvi alilipasua?
(ii) Samaki aliposema "Mtaka yote kwa pupa hukosa yote" alikuwa na maana gani?
(iii) Taja tabia mbili za mvuvi
(iv) Eleza maana ya maneno yafuatayo kama yalivyotumika katika habari uliyosoma.
  (a) Ufukwe
  (b) Rairai
  (c) Ghadhabu
  (d) Hayawani
  (e) Bashasha
(v) Andika kichwa cha habari cha hadithi uliyosoma.

2. Fupisha habari uliyosoma kwa maneno yasiyopungua mia moa (100) na yasiyozidi maneno mia moja na arobaini (140).

SEHEMU B: (ALAMA 25)
SARUFI NA UTUMIZI WA LUGHA
Jibu maswali YOTE katika sehemu hii

3. Eleza maana ya istlahi zifuatazo kwa kutumia mfano mmoja kwa kila istilahi.
  (a) Silabi (b) Mofimu (c) Shina
  (d) Rejesta (e) Mzizi

4. Changanua sentensi zifuatazo kwa njia ya ngoe kwa kutumia istilahi za kikazi.
  (a) Mwizi aliyetusumbua sana amekamatwa leo asubuhi.
  (b) Mama alikuwa anataka kwenda kupika chakula chake.

5. Eleza dhima za mofimu zilizopigiwa mstari katika maneno yafuatayo:
  (a) Anasikiliza (b) Aliyesikia (c) Atasoma
  (d) Aliyethubutu (e) Nimejikata

6. "Upatanisho wa kisarufi ni kigezo kimojawapo kati ya vigezo vya kuunda ngeli za nomino". Thibitisha dai hili kwa kutunga sentensi mbili katika kila ngeli, kwenye ngeli zifuatazo:
  (a) Ngeli ya pili (b) Ngeli ya tatu (c) Ngeli ya sita
  (d) Ngeli ya tano (e) Ngeli ya saba

7. Taja na fafanua njia tano za uundaji wa msamiati katika lugha ya Kiswahili. Toa mifano miwili katika kila njia utakayoitaja.

SEHEMU C (ALAMA 10)
UANDISHI
Jibu swali **moja (1)** kutoka katika sehemu hii.

8. Ukiwa kama mkuu wa shule, andika barua ya mwaliko kwa Mkuu wa Wilaya ya Masasi S.L.P 2993 Mtwara ili aje kukabidhi vyeti kwenye sherehe ya kuhitimu Kidato cha nne. Jina lako liwe Elimu Mbele.

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9. Andika mfano wa kumbukumbu ya mkutano unaofanyika kwa mara ya kwanza kuhusu mahafali ya kidato cha nne katika shule yako.

SEHEMU D (ALAMA 10)
MAENDELEO YA KISWAHILI
Jibu swali la kumi (10).

10. Kwa kutumia hoja nne (4) Onyesha juhudi za Waingereza katika kukuza Lugha ya Kiswahili nchini wakati wa utawala wao.

SEHEMU E (ALAMA 45)
FASIHI KWA UJUMLA
Jibu maswali matatu (3) kutoka katika sehemu hii. Swali la 15 ni la lazima.

ORODHA YA VITABU

USHAIRI

Wasakatonge	-	MUHAMMED S. KHATIB (DUP)
Malenga Wapya	-	TAKILUKI (DUP)
Mashairi ya Chekacheka	-	THEOBALD A. MVUNGI (EP & D. LTD)

RIWAYA

Takadini	-	BEN J. HANSON
Watoto wa Mama ntilie	-	E. MBOGO
Joka la Mdimu	-	A. SAFARI

TAMTHILIYA

Orodha	-	STEVE REYNOLDS (MA)
Ngoswe Penzi Kitovu cha Uzembe	-	E. SEMZABA (ESC)
Kilio Chetu	-	MEDICAL AID FOUNDATION (TPH)

11. Jadili mambo ya kimsingi yanayodhihirisha uhai na uhalisia wa fasihi simulizi.

12. "Mfumo wa maisha ndio unaweza kusababisha kutokea kwa kero mbalimbali katika jamii". Kwa kutumia riwaya mbili ulizosoma thibitisha kauli hii kwa kutoa kero tatu (3) kutoka kila riwaya.

13. Washairi ni watu mabingwa waliobobea katika kutumia taswira zinazotoa ujumbe kwa jamii. Tumia taswira tatu kwa kila diwani mbili zilizoorodheshwa hapo juu kuthibitisha kauli hii.

14. Waandishi wa tamthiliya wamemchora mwanamke katika sura tofauti tofauti. Thibitisha kauli hii kwa kutumia waandishi wawili (2) wa tamthiliya mbili ulizosoma. Toa hoja tatu kwa kila tamthiliya.

15. Tunga tamthiliya fupi kuhusu umuhimu wa katiba kwa watanzania wa leo. Tamthiliya hiyo isipungue maneno mia tatu (300).

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DODOMA REGION
REGIONAL FORM IV MOCK EXAMINATION – MAY 2019

032/1

CHEMISTRY - 1

TIME: 3 Hours **Wednesday, 8th May 2019 a.m.**

INSTRUCTIONS

1. This paper consists of sections A, B and C.
2. Answer **all** questions in this paper.
3. Write your examination number on every page of your answer sheet(s).
4. Calculators and cellular phones are **not** allowed in the examination room.
5. The following constants may be used:

   Atomic masses:
   H = 1, Li = 7, C = 12, O = 16, Na = 23, Al = 26, Cl = 35.5, K = 39, Cu = 63.5

   Avogadro's number = 6.02×10^{23}.

   GMV at s.t.p. = 22.4 dm^3.

   1 Faraday = 96,500 coulombs.

   Standard pressure = 760 mm Hg.

   Standard temperature = 273 K.

   1 litre = 1 dm^3 = 1000 cm^3.

This paper consists of 6 printed pages

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SECTION A (20 MARKS)

Answer ALL questions in this section

1. For each of the following items (i) – (x), choose the correct answer from the given alternatives and write its letter besides the item number in the answer sheet provided:

(i) A Chemist should acquire the following skills except:

A. Problem identification
B. Experimentation
C. Observation
D. Data analysis
E. Surgery

( )

(ii) The valency of an element with an atomic number of 8 is:

A. 0 B. 1
C. 2 D. 3
E. 4

( )

(iii) Which of the following warning signs in likely to appear on the bottle containing caustic soda in the laboratory

A. Oxidant B. Corrosive
C. Toxic D. Harmful
E. Explosive

( )

(iv) ..... is the only alkaline gas.

A. Hydrogen chloride
B. Nitrogen dioxide
C. Chlorine
D. Sulphur-dioxide
E. Ammonia

( )

(v) A steady current of 4 amperes was passed through an aqueous solution of copper II Sulphate for 1800 seconds. The mass of copper deposited was:

A. 63.5g B. 31.75g
C. 1.18g D. 2.38g
E. 11.8g

( )

(vi) When chlorine gas is passed through a green solution of iron II chloride, the solution changes to yellowish brown because

A. Fe^{2+} were oxidized to Fe^{3+}
B. Fe^{3+}were reduced to Fe^{2+}
C. Fe^{2+}were reduced to Fe^{3+}
D. Fe^{3+}were oxidized to Fe^{2+}
E. Fe^{3+}were neither reduced nor oxidize

( )

(vii) A gas burned in air forms carbon dioxide and water only. From this experiment, the gas is likely to be:

A. Hydrogen B. Carbon-dioxide
C. Ethane D. Nitrogen
E. Ozone

( )

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SECTION B (54 MARKS)

Answer ALL questions from this section

3. (a) Study the following part of the periodic table with some elements represented by letters and answer the questions that follow:

I	II	III	IV	V	VI	VII	VIII
						B	
	A				C	D	

(i) State how electro negativity varies from A to C and from B to D.

(ii) Write the electronic configuration of A, B, C^{2-} and D.

(b) Briefly explain what will happen when;

(i) Concentrated sulphuric acid is exposed to the atmosphere?

(ii) Iron (II) sulphate is exposed to air for a long time?

(iii) A bottle containing $AgNO_3$ is left open?

4. (a) Name the following compounds:

(i) An amphoteric oxide which is yellow when hot and white when cold.

(ii) A white precipitate which is formed when aqueous silver nitrate is added to aqueous sodium chloride.

(b) By using a balanced chemical equation explain what will you observe when a steady stream of chlorine passed to the following;

(i) Overheated white phosphorous

(ii) Over heated iron wire

(iii) Through potassium iodide solution

5. (a) List any Six (6) Characteristics of a good fuel.

(b) An element M consists of three isotopes with mass numbers 22, 24 and 25 with Percentage abundances of 89.6%, 6.4% and 4.0% respectively. Find the relative Molecular mass of element M.

(c) Suggest one best method for separating each of the following mixtures:

(i) Common salt and water

(ii) Iodine and sand.

(iii) Pieces of iron and sand.

6. (a) Define the following terms

(i) Molar solution

(ii) A standard solution

(b) $25cm^3$ of 0.1M Na_2CO_3 neutralizes a solution containing 2.5g sulphuric acid in 250 cm^3 of solution. Calculate

(i) The molarity of sulphuric acid solution

(ii) The volume of acid used.

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(viii) A salt which on exposure to air becomes covered with a liquid is:
A. Efflorescent
B. Deliquescent
C. Hygroscopic
D. Hydrolytic ( )
E. Vaporization

(ix) A less electropositive metal may be extracted by:
A. Ore concentration and electrolytic reduction.
B. Ore concentration and chemical reduction.
C. Ore concentration and purification
D. Ore concentration, reduction and purification. ( )
E. Ore concentration ,purification and roasting

(x) The rate of a chemical reaction can be accurately measured by measuring the rate of
A. Change of colour.
B. Change of temperature of the reactants.
C. Product formation.
D. Change of state of reactants. ( )
E. Change of state of the product.

2. Match the items in list **A** with the responses in list **B** by writing the letter of the correct response beside the item number in the table provided.

LIST A	LIST B
(i) Molecular formular	A. Determines the chemical properties of a certain class of organic compounds.
(ii) Hydrogenation of alkenes	B. Group of atoms present in a compound and are charged.
(iii) Used in fastening the ripening of fruits	C. Normal butane and 2-methylpropane.
(iv) Esterification.	D. Ethene
(v) Butanoic acid	E. Ethyne
(vi) Prepared by the reaction between water and calcium carbide	F. Converting them to Alkanes.
(vii) Isomers of C_4H_{10}	G. Making of soap from oil or fat.
(viii) Functional group	H. Ester formation
(ix) Saponification	I. Reaction between acid and base to form ester and water only
(x) Carbon tetrachloride.	J. C_3H_7COOH
	K. C_4H_9COOH
	L. Halo alkane
	M. Methane
	N. Shows the actual number and type of atom present in a molecule of a compound.
	O. Show the composition of each element by mass

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7. (a) State two (02) uses of each of the following:
   (i) Ethanoic acid.
   (ii) Carbon-dioxide gas
   (b) Write the structural formula of the following
   (i) 2 - butene
   (ii) 2 - pentyne
   (iii) 1, 2 - dichloroethane
   (iv) 2, 4 - dimethylhexane.

8. (a) (i) State Le Chatelier's principle
   (ii) Write down three factors affecting the equilibrium reactions.
   (b) The reaction which produces methanol from carbon monoxide and hydrogen is represented by the equation $CO_{(g)} + 2H_{(g)} \rightleftharpoons CH_3OH_{(g)}$ $\Delta H = -94kJmol^{-1}$. The reaction is carried out at high pressure to give a good yield of methanol.
   (i) Explain why increase in pressure gives a better yield of methanol
   (ii) The value of ΔH is negative. What does this tell about the reaction?
   (iii) With a reason, state whether a high temperature or low temperature will give a better yield of methanol.

9. (a) Calculate the percentage by composition of the underlined elements in the following compounds:
   (i) $H_2\underline{S}O_4$
   (ii) $\underline{Cu}Cl_2$
   (b) Name the gas produced when:
   (i) Potassium chlorate decomposes
   (ii) Zinc granules react with dilute hydrochloric acid.

10. (a) State Faraday's first law of electrolysis
   (b) During electrolysis of an aqueous solution of a salt of metal M, a current of 2.0A was passed for 32 minute and 10 seconds. The mass of M deposited was 2.24g and relative atomic mass of metal M is 112.
   (i) On which electrode was the metal deposited?
   (ii) Calculate the amount of charge needed to deposit 1 mole of metal M in coulomb
   (iii) Calculate the charge carried on the ion.
   (iv) Write an ionic equation to show how the ions of metal M are discharged at the electrode.

11. (a) Differentiate Diamond from graphite (four points).
   (b) Explain why Diamond does not conduct electricity while graphite does
   (c) State any two uses of graphite.

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SECTION C: (26 MARKS)

Answer all questions in this section.

12. (a) What is terrestrial pollution?
    (b) Explain five (5) causes of terrestrial pollution
    (c) Mention three (3) hazards of terrestrial pollution
    (d) List 4 methods of preventing terrestrial pollution.

13. You are at Oldoinyo Lengai village in Arusha region where there is potential sulphur deposit. The village development society aims to extract the sulphur and has come to you for advice. Explain to the society
    (a) The equipment needed for the extraction of sulphur use an illustration to make the explanation clear.
    (b) The process involved during and after the extraction
    (c) The effect of extraction on the environment
    (d) Commercial use of sulphur

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THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

DODOMA REGION
REGIONAL FORM IV MOCK EXAMINATION – MAY 2019

012 **HISTORY**

TIME: 3 Hours **Tuesday, 7th May 2019, p.m.**

INTRODUCTIONS

1. This paper consists of sections **A**, **B** and **C**
2. Answer all questions in sections **A** and **B** and tree questions from section **C**
3. Cellular phones **are not allowed** in the examination room.
4. All drawings **should be in a pencil**
5. Write **your examination numbers** on every page of your answers sheet (s)

This paper consists of **6** printed pages

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SECTION A (20 MARKS)
ANSWER ALL QUESTIONS IN THIS SECTION.

1. For each of the items (i - x) choose the most correct answer among the given alternatives and write its letter beside the item number in the answer booklet provided.

(i) The appearance of both Zinjanthropus and Homo habilis marked the beginning of a period in man's history known as:-
- (a) Neolithic age
- (b) Modernization Age
- (c) Early Stone Age
- (d) Middle Stone Age.
- (e) Pre-colonial period.

(ii) In areas such as Kondoa Irangi, painting and drawing in caves give evidence of the activity of
- (a) Education for adaptation
- (b) Bantu speaking people
- (c) Emergence of slave trade
- (d) Settled communities
- (e) Iron smelters

(iii) People whose work is to study and explain remain which shows man's physical development and tools he made and used are called
- (a) Archaeologist
- (b) Instructor
- (c) Iron smiths
- (d) Evolutionist
- (e) Evolutionist

(iv) The Trans-Atlantic slave trade had the following outcome except:-
- (a) Loss of African culture
- (b) Loss of African technology.
- (c) Loss of manpower in Africa
- (d) The growth of forest states of Benin, Oyo and Ife.
- (e) Scramble of Africa by European nations.

(v) One of the features of Monopoly Capitalism is:-
- (a) Merging bank capital and trade
- (b) Formation of big African monopolies
- (c) Expansion of industrial production in the capitalist countries
- (d) Growth of small scale production in Europe
- (e) Emergence of slave trade and slavery in Africa

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(ii)	Spiritual struggle to defend religious faith.	D. First world war
(iii)	The war fought between the Xhosa against the Boers due to the question of land.	E. Majimaji war
(iv)	Ndebele-Shona uprising against the British South African company in Southern Rhodesia.	F. Jihad war
(v)	France lost two provinces after being defeated by German in 1871.	G. Wars of Dispossession
(vi)	It was fought in Nigeria from 1967 to 1970.	H. Civil war
(vii)	The name given to the situation during which there was a tension between the Eastern and Western block.	I. Mfecane war
(viii)	The struggle between European powers which lasted in 1918.	J. Ghana – Togo war
(ix)	It caused the migration of the Ngoni during the 19th century.	K. Biafra war
(x)	The name given to the situation where there is political unrest within the some country motivated by either ethnical struggle or rebels.	L. Chimurenga war
		M. Franco – Prussian war
		N. Uganda civil war
		O. Cold war

SECTION B (20 MARKS)

Answer all questions in this section.

3. (a) Draw a sketch map of Africa and locate by using roman numbers:
   (i) Uncolonized country which was found primarily to freed slaves from U.S.A
   (ii) A Country whose independence sharpened the continent during struggle for independence.
   (iii) A Country which waged Mau-Mau war as the desire to liberate from foreign rule.
   (iv) The trusteeship colony under the British from 1945 to 1961
   (v) The country in which the 1994 genocide took place

   (b) Name two forms of feudal system and three feudal relations which developed in East Africa before the onset of colonialism.

4. (a) Arrange the following statements in chronological order by writing number 1-5 beside the item number.
   (i) In December 1963 the British handed political power to the minority
   (ii) It was soon followed by the union between Tanganyika and Zanzibar forming the United Republic of Tanzania.
   (iii) The 1964 Zanzibar revolution was the second attempt made to bring about real political independence.

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(vi) One of the major problems that hindered the attainment of unity in Uganda immediately after independence was:-
   (a) Religious Conflicts
   (b) Dictatorship of Idd Amin
   (c) Civil Wars
   (d) Racial Segregation
   (e) Tribalism

(vii) The Hamerton Treat (1845) was about :-
   (a) Allowing Sultan of Zanzibar to buy slaves outside East Africa
   (b) Forbidding Sultan of Zanzibar to export slaves outside East Africa
   (c) Allowing Sultan of Zanzibar to sell slaves in Arabia
   (d) Abolishing Slave trade and Slavery in all British Colonies
   (e) Closing all Slave markets along the coast of East Africa

(viii) During the colonial period the provision of social services was not given priority to Africans because:-
   (a) Arabs and Portuguese had provided social services before colonialism.
   (b) Africans refused modernity
   (c) Africans had all social services.
   (d) Social services were part of profit to European capitalist.
   (e) Social services were not part of profit to European capitalists.

(ix) The British exercised direct rule policy in some parts of Zimbabwe because:-
   (a) It was expensive to administer
   (b) The natives and Traditional chiefs opposed colonial rule
   (c) Zimbabwe was dominated by centralized states.
   (d) Zimbabwe had enough experienced personnel.
   (e) The British were afraid of eliminating the local administrative set up.

(x) The type of colonial Agriculture which predominated in Kenya was:-
   (a) Plantation
   (b) Peasant
   (c) Co-operative
   (d) Settler
   (e) Pastoralism.

Match the items in **list A** with the correct responses in **list B** writing the letter of correct response beside the item number in the answer sheet provided

LIST A	LIST B
(i) African warriors were encouraged by some magic water which would prevent bullets from penetrating their bodies.	A. Second world war B. Tanzania – Uganda war C. War against terrorism

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(iv) This event was an important step towards the achievement of East African Unity.

(v) The revolution was intended to transfer power to the majority.

(b) The Items (i – v) consist of four sentences which form a complete historical meaning. One of the sentences (A - E) is missing, identify the missing sentence and write its letter beside the item number.

(i) 1. Clan organization was formed by several related families.
2. .....
3. Clan organization developed both matrilineal and patrilineal societies
4. In matrilineal societies clan heritage was based on the mother.

A. People in clan organization were governed by grand fathers
B. People in clan organization were governed by clan heads
C. People in clan organization were governed by the kings
D. People in clan organization were governed by chiefs
E. People in clan organization were governed by queens.

(ii) 1. The important items from West Africa to North Africa were gold and ivory.
2. West Africa obtained salt, dates and horses from North Africa.
3. .....
4. It expanded after the introduction of camel during the 9th century

A. This trade took place through the Atlantic Ocean
B. This trade took place across the Saharan Desert
C. This trade took place across the Indian Ocean
D. This trade took place after the partition of Africa
E. This trade took place after the decolonization of Africa.

(iii) 1. After the Second World War the United States of America advocated for decolonization.
2. .....
3. The United States of America wanted the colonizing power to give independence to their colonies.
4. She wanted to have a freer hand in the exploitation of the ex – colonies.

A. This was known as the Open Door policy
B. This was known as the Veto policy
C. This was known as the Assimilation policy
D. This was known as the indirect rule policy
E. This was known as the direct rule policy

(iv) 1. Indirect rule was introduced in Africa by Fredrick Lugard
2. .....
3. The system was cheap
4. The system reduced resistance from the natives.

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A. Africans used Jumbes and Akidas to represent their problems
B. France applied the system in her colonies in Africa
C. The system improved the living standard of the Africans
D. The system helped the British to consolidate tribalism.
E. The British preferred direct rule due to its efficiency.

(v) 1. The First World War lasted from 1914 to 1918
2. It was primarily a struggle between the major powers of Europe.
3. .....
4. Although the First World War involved purely European issues. It was inevitable that the overseas empires of two sides would be involved.
   A. Britain France and their allies were fighting against Germany Austria and Turkey
   B. The killing of Archduke Franz Ferdinand became one of the long term causes of war.
   C. Germans, France and Italy joined and fight against triple alliance of British, Russia and U.S.A.
   D. Tanganyika was to be under Germany rule after the war.
   E. All European Nations struggled for colonies in the Balkan states.

SECTION C (60 MARKS)
ANSWER THREE QUESTIONS FROM THIS SECTION

5. Explain six social factors for early interaction in pre-colonial Africa.
6. What were the main characteristics of centralized state in Pre-Colonial Africa? (give six points)
7. Why British use indirect rule system and not direct rule. (give six points)
8. With vivid examples, explain six main causes of the First World War.
9. Critically examine six effects of colonial education in East Africa.
10. Analyze the problems experienced during the struggle for independence in Tanganyika.

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